Ruitong Tian

ruitongtian@ufl.edu | Website | Google Scholar |

EDUCATION

- **University of Florida** — *2025 – Present*
  Ph.D. student | Sustainable Construction. Advisor: Eric Jing Du. — Gainesville, FL
  Research focus: Dexterous manipulation, robot learning, and tactile sensing.
- **National University of Singapore** — *2023 – 2025*
  M.S. & Year 4 Undergraduate Exchange | Infrastructure Digitalization & Management. — Singapore
  Advisor: Prof. Poh Leong Hien.
- **South China University of Technology** — *2020 – 2024*
  B.E. | Green Intelligent Construction. Advisor: Prof. Nan Hu. — Guangzhou, China

PROJECTS & PUBLICATIONS

ExoBridge: Learning a Bare Hand to Hand-Worn Exoskeleton Mapping through Human Limb Coupling — **Under submission**, *University of Florida*

Ruitong Tian, Xianyao Li, Noah B. Wilson, Fang Xu, Eric Jing Du.

- Developed bare-hand RGB-to-exoskeleton mapping with MediaPipe hand/scene crops, frozen DINOv2 features, and a causal temporal model to predict motion, contact, and tactile intensity.
- Evaluated 1,215 demonstrations from 31 sessions across four tasks: contact AUROC 0.916, intensity correlation 0.790, and 44.7% lower motion error versus a zero-change reference for primary channels.

From Wearable Interfaces to Dexterous Policies: Contact Shifts and Tactile Representations — **Under submission**, *University of Florida*

Ruitong Tian, Fang Xu, Noah B. Wilson, Xianyao Li, Eric Jing Du.

- Studied wearable-interface geometry and tactile inputs for Diffusion Policy manipulation with a DINO encoder, temporal U-Net, 12-DoF XHand, and Franka Emika arm.
- Tested 12 policy–task conditions with 20 rollouts each across four tasks; spatial contact and force inputs increased pooled success from 35% to 52% versus binary contact.

Semantically Enhanced Eye-Tracking Controls in Human-Robot Interaction: A Comparison Study — **Ergonomics**, **Accepted Oct. 9, 2026**, *University of Florida*

Ruitong Tian, Hengxu You, Fang Xu, Jiahao Wu, Bowen Sun, Thanakon Uthai, Eric Jing Du.

- Co-designed a 30-participant gaze/voice/gesture study using HoloLens 2, Franka Emika, FastSAM, and RGB-D localization; curated and analyzed task performance, gaze behavior, and NASA-TLX workload.
- Measured mean step completion times of 17.10 s with gaze versus 32.78 s with hand gestures; gaze analysis used a quality-screened subset of 19 participants and 96 trials.

Nudge Before You Push: Physics-Aware Navigation via Tactile Probing — **Under submission**, *University of Florida*

Xianyao Li, Fang Xu, **Ruitong Tian**, Bowen Sun, Xiao Hu, Yang Ye, Jing Du.

- Coauthored TANav tactile-probing navigation; simulation missed-push rate was 5.5% versus 28.7% with mean peak-force thresholds at comparable low-risk operating points.

Controlling crack propagation in layered beams with architected lattice-reinforced composite interlayer designs — **Construction and Building Materials (2024)**, *South China University of Technology*

Binglin Xie, **Ruitong Tian**, Haiyang Zhao, Tongtong Ye, Yuan Zhang, Nan Hu.

- Designed Rhino/Grasshopper lattices and FDM prototypes; validated Abaqus crack models using fracture/fatigue tests and digital image correlation.

ADDITIONAL RESEARCH EXPERIENCE

Machine Learning for Thermal-Image Damage Prediction — *National University of Singapore*

- Developed CNN/knowledge-transfer models in PyTorch (prediction correlation 0.98); automated 2,000+ Abaqus simulation samples and processed 150+ thermal images with Python/MATLAB.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, MATLAB, Fortran.
- **Robot Learning & ML:** PyTorch, Hugging Face, imitation learning, Diffusion Policy, CNNs, temporal modeling.
- **Computer Vision & Data:** OpenCV, DINO/DINOv2, MediaPipe, FastSAM, NumPy, pandas, SciPy, scikit-learn.
- **Robotics & Hardware:** ROS/ROS 2, dexterous manipulation, tactile sensing, DexUMI, XHand, Franka Emika, HoloLens 2.
- **Systems & Compute:** Linux (Ubuntu), Git, Slurm; multi-node GPU training (HiPerGator, PyTorch/CUDA).
- **Simulation & Design:** MuJoCo, Isaac Sim, Abaqus, Rhino/Grasshopper, 3D printing, digital image correlation.